

SATHWIK VALLALA

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Holding H1-B valid till Oct 2028 and an approved i-140

PROFESSIONAL SUMMARY

Application and Product Security Engineer with 6+ years of experience securing software across the Secure Software Development Lifecycle (Secure SDLC), from threat modeling and secure-by-design reviews through secure code review, CI/CD security, vulnerability management, penetration testing, and incident response. Hands-on experience embedding SAST, DAST, and SCA into DevSecOps workflows; assessing API, Kubernetes, and container security; and building Python security automation. Applies agentic AI and LLM Security techniques to accelerate analysis and remediation, brings practical AWS and cloud security knowledge, and partners with engineering teams to make secure implementation the default. AI Security certified.

PROFESSIONAL EXPERIENCE

TryHackMe

August 2026 - September 2026

AI Security

Remote

- Applied knowledge from TryHackMe's AI Security path and independent research to develop an open-source AI Attack & Defence Map.
- Studied the AI lifecycle, from training and fine-tuning through deployment and monitoring, and organized 58 attack techniques with approximately 174 defence entries.
- Learned how training-data poisoning, model backdoors and compromised dependencies can affect AI systems; documented their mechanisms, impacts and mitigations.
- Examined direct and indirect prompt injection, jailbreaking and insecure output handling in LLM applications.
- Explored RAG security risks, including corpus poisoning, vector database abuse and sensitive information disclosure.
- Investigated agent security threats involving tool misuse, malicious MCP integrations, memory poisoning and credential exposure.
- Applied MITRE ATLAS and OWASP LLM Top 10 references to connect relevant threats with established security frameworks.
- Built interactive visualizations showing attack progression and where defensive controls can intervene.
- Translated technical research into accessible explanations, curated resources and contribution guidelines for community learning.

Omniassa

June 2024 - July 2026

Application & Product Security Engineer

Remote

- Owned Application Security and Product Security risk across 30+ customer features, using OWASP-aligned vulnerability assessments and attack-surface analysis to prioritize threats and recommend practical defenses.
- Assisted 20+ releases by incorporating secure code review and high-confidence SAST, DAST, and SCA findings into CI/CD security workflows.
- Led risk-based vulnerability management for 50+ critical findings, applying CVSS to prioritize remediation and validate practical mitigation strategies.
- Applied AI Security, agentic/LLM threat analysis, and AI-assisted penetration testing across 10+ product releases, accelerating reconnaissance, payload generation, and vulnerability triage.
- Expanded AI-assisted static analysis and code understanding across 75% of product codebases, identifying risks earlier and scaling secure code review coverage.
- Presented complex security risks to cross-functional teams and led a monthly cybersecurity roundtable on emerging threats, strengthening secure-development awareness.

Broadcom

November 2023 - June 2024

Product Security Engineer (CarbonBlack)

Herndon, VA

- Identified 10 Product Security vulnerabilities through vulnerability assessment and CVSS-based prioritization, helping engineering teams reduce attack surface and focus remediation on material risk.
- Validated application vulnerabilities through OWASP-aligned penetration testing and DAST across 4 product releases.
- Conducted STRIDE threat modeling and early security architecture reviews to guide pragmatic secure-by-design decisions with engineering teams.
- Performed SAST and SCA informed secure code reviews within CI/CD workflows to identify and mitigate exploitable software and dependency risk.
- Translated complex technical findings into clear risk context, building understanding and remediation buy-in across product teams.

VMware

April 2022 - November 2023

Security Response Engineer

Herndon, VA

- Owned the vulnerability-management lifecycle for 50+ externally reported findings, performing triage, risk prioritization, remediation coordination, and closure across product teams.
- Conducted vulnerability assessments, assigned CVSS v3.1 severity scores, and documented exploitation paths, mitigations, and compensating controls.
- Partnered with security researchers on responsible disclosure and published 10+ VMware Security Advisories (VMSAs) with clear customer remediation guidance.
- Monitored active exploitation, developed indicators of compromise (IOCs), and updated mitigations and advisories as threat conditions evolved.
- Developed security-response automation that streamlined vulnerability triage, stakeholder coordination, and operational workflows.

ADF Solutions

December 2020 - March 2022

Digital Forensic Researcher & Developer

Reston, VA

- Built 10+ production Python plugins to parse and extract digital artifacts, automating evidence collection and investigative workflows.
- Designed collection approaches for 20+ digital artifacts and independently developed new modules for complex evidence-acquisition use cases.
- Improved module accuracy and reliability through code-level testing, quality assurance, and validation before integration into the operational DEI Tool.

SOC Experts

July 2019 - October 2019

Security Analyst

Bangalore, India

- Enhanced security operations through hands-on use of Splunk, QRadar, and LogRhythm for centralized monitoring, threat detection, and coordinated incident response.
- Analyzed and monitored security logs to identify suspicious activity, uncover potential incidents, and support timely threat detection across simulated SOC environments.
- Collected and analyzed digital evidence using basic forensic techniques, enabling accurate investigative findings and supporting effective incident response.
- Built simulated SOC response capabilities by collaborating on threat detection, investigation, and remediation workflows, reinforcing effective security operations and team coordination.

Berry9 IT Services Pvt Ltd

June 2017 - June 2018

Security Analyst

Hyderabad, India

- Established practical ethical hacking expertise through various projects, applying offensive security methodologies to uncover vulnerabilities and advance penetration testing capabilities.
- Identified vulnerabilities across real-world healthcare penetration testing environments, applying offensive security methodologies to expose weaknesses and strengthen organizational risk awareness.
- Achieved Certified Ethical Hacker (CEH) certification by demonstrating proficiency in ethical hacking principles, vulnerability assessment, and penetration testing techniques.

EDUCATION

George Mason University

Master's, Digital Forensics & Cyber Analysis

GPA: 3.89

Jawaharlal Nehru Technological University

Bachelor's, Cyber Security

GPA: 4.0

CERTIFICATIONS

- AI Security | TryHackMe | September 2026
- Security Code Review Ninja - 2nd Degree Black Belt | Secure Coding Dojo | December 2023
- Digital Evidence Investigator | ADF Solutions | December 2020
- Certified Autopsy User | Basis Technology | October 2020
- Certified Ethical Hacker | EC Council | October 2017

SKILLS

Security: Application security, Product security, Secure SDLC, Secure code review, Security design review, STRIDE, Threat modeling, Vulnerability Analysis, Penetration Testing, API Security, Attack surface analysis, OWASP, Incident Response, AI Security, AI Assisted Security

SecOps & Automation: SAST, DAST, Coverity, Semgrep, CI/CD Security, Shift left, SCA, SIEM, DevSecOps, MLOps

Platform & Detection: AWS, Cloud security, Kubernetes, Container security, Docker, SaaS, Linux/Unix, Splunk, QRadar, LogRhythm, Log analysis, Security monitoring, Digital forensics, Network forensics, Memory forensics, Reverse Engineering, OSINT, Ghidra, Autopsy, Volatility, EnCase, FTK

Tools: Python, Bash, PHP, SQL, Burpsuite, OWASP ZAP, Metasploit, Nmap, Nessus, Wireshark, SQLMap, Kali Linux

PROJECTS & OUTSIDE EXPERIENCE

Collaborative ToDo - [Link to project](#)

March 2026 - April 2026

- Built and deployed a collaborative task-management application using AI-assisted software-development workflows and self-hosted infrastructure.
- Implemented secure passwordless authentication and access control to protect shared user data while streamlining access.

Autopsy Whatsapp Plugin - [Link to project](#)

April 2020 - May 2020

- Enhanced Autopsy capabilities by independently developing a new plugin, demonstrating expertise in forensics tool customization & automation.
- Recovered & parsed digital evidence from the WhatsApp application on Windows systems, utilizing advanced data extraction & parsing techniques.

Chat over network with Python - [Link to project](#)

June 2018 - July 2018

- Built a privacy-forward chat application with strong cryptography, empowering secure communication.
- Implemented robust end-to-end encryption for secure communication, ensuring message confidentiality & protecting user data
- Enabled user control & independence by designing a self-hostable chat server, fostering privacy & data sovereignty

File Sharing - [Link to project](#)

March 2018 - April 2018

- Streamlined internal file sharing with a user-friendly platform designed for specific network needs.
- Prioritized security by implementing password authentication to restrict unauthorized access to shared files.